

RESEARCH PRESENTATION

Green Physics Approach to Multi-functional Glass

Experimental Evaluation of Transparency and Ionizing Radiation Shielding

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CHAPTER ONE

Introduction

Background, motivation, and research objectives for developing green multi-functional glass for ionizing radiation shielding applications.

Research Background & Motivation

The Radiation Challenge

Ionizing radiation (gamma rays, X-rays) poses significant health risks across medical, industrial, and nuclear environments. Effective shielding is essential for personnel protection in:

- Medical imaging (PET, CT, radiotherapy)
- Nuclear power plants and research facilities
- Industrial radiography and hot cells

Limitations of Current Materials

Lead (Pb): High toxicity, environmental hazard, poor mechanical strength, oxidation concerns. Many facilities now require glove handling.

Concrete: Opaque (no visual monitoring), bulky, fractures under prolonged radiation exposure, water evaporation at high energies degrades performance.

The Green Physics Solution

Glass offers a unique combination of transparency, versatility, and radiation attenuation capability. The "Green Physics" approach emphasizes:

- Eco-friendly, lead-free compositions using heavy metal oxides (Bi_2O_3 , BaO , WO_3 , Gd_2O_3)
- Sustainable synthesis using recycled soda-lime-silica (SLS) glass waste
- High effective atomic number (Z_{eff}) with maintained optical transparency

Problem Statement & Research Objectives

 **Key Problem:** Existing radiation shielding solutions force a trade-off between **transparency** and **protection efficiency**. No current material optimally satisfies both requirements while being environmentally sustainable.

Research Objectives

- 1

Develop lead-free glass compositions with high radiation attenuation using eco-friendly heavy metal oxides
- 2

Maintain optical transparency in the visible light range for practical viewing-window applications
- 3

Characterize nuclear shielding parameters (LAC, HVL, MFP, Z_{eff}) across photon energy ranges
- 4

Validate theoretical predictions through experimental measurements and computational simulations

TARGET APPLICATIONS

- Medical Imaging (PET, CT scans)
- Nuclear Power Plants
- Radiation Therapy Rooms
- Hot Cells & Radioisotope Handling
- Industrial Radiography
- Research Reactors

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CHAPTER TWO

Materials and Methods

Glass composition design, melt-quenching synthesis methodology, and sample preparation procedures.

Glass Composition Design

Base System: Soda-Lime-Silica (SLS) + Heavy Metal Oxides

Oxide (mol%)	S1	S2	S3	S4	S5	S6
SiO ₂	60	55	50	45	40	35
Na ₂ O	15	15	15	15	15	15
CaO	10	10	10	10	10	10
Bi ₂ O ₃	5	10	15	20	25	30
BaO	10	10	10	10	10	10
Density (g/cm ³)	3.42	3.78	4.15	4.52	4.89	5.26

Table 1: Glass composition series with increasing Bi₂O₃ content

Key Additives & Roles

- Bi₂O₃** (Z=83): Highest atomic number element; enhances photoelectric absorption at low energies
- BaO** (Z=56): Increases density and refractive index; improves overall attenuation
- WO₃** (Z=74): High density; strengthens glass network structure
- TeO₂**: Improves thermal stability and optical transmission properties
- Gd₂O₃** (Z=64): Excellent neutron absorption; highest capture cross-section

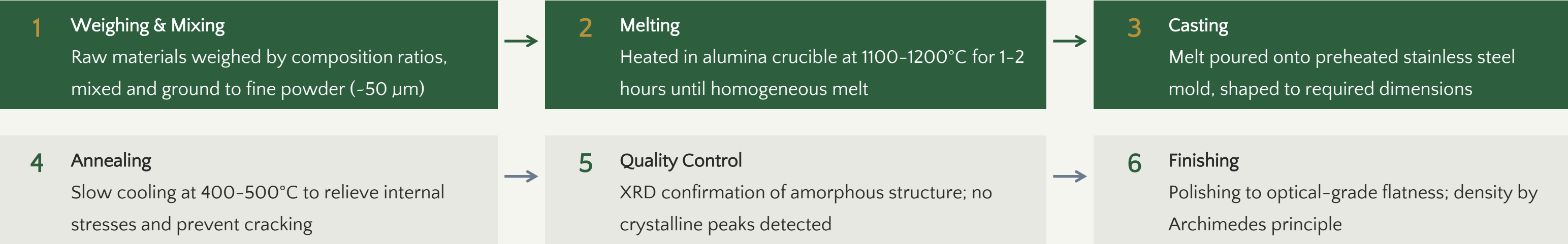
Design Rationale

The composition design follows a systematic approach to maximize the **effective atomic number (Z_{eff})** while preserving optical transparency:

- Gradual substitution of SiO₂ with Bi₂O₃ to increase Z_{eff} and density
- Constant BaO content to maintain network stability and refractive index
- SLS waste glass as sustainable base material — reduces environmental impact

Synthesis & Sample Preparation

Method: Melt-Quenching Technique



Processing Parameters

Parameter	Value / Condition
Melting Temperature	1100 - 1200°C
Holding Time	1 - 2 hours
Crucible Material	Alumina (Al ₂ O ₃)
Annealing Temperature	400 - 500°C
Cooling Rate	Gradual (furnace cooling)
Powder Particle Size	~ 50 μm

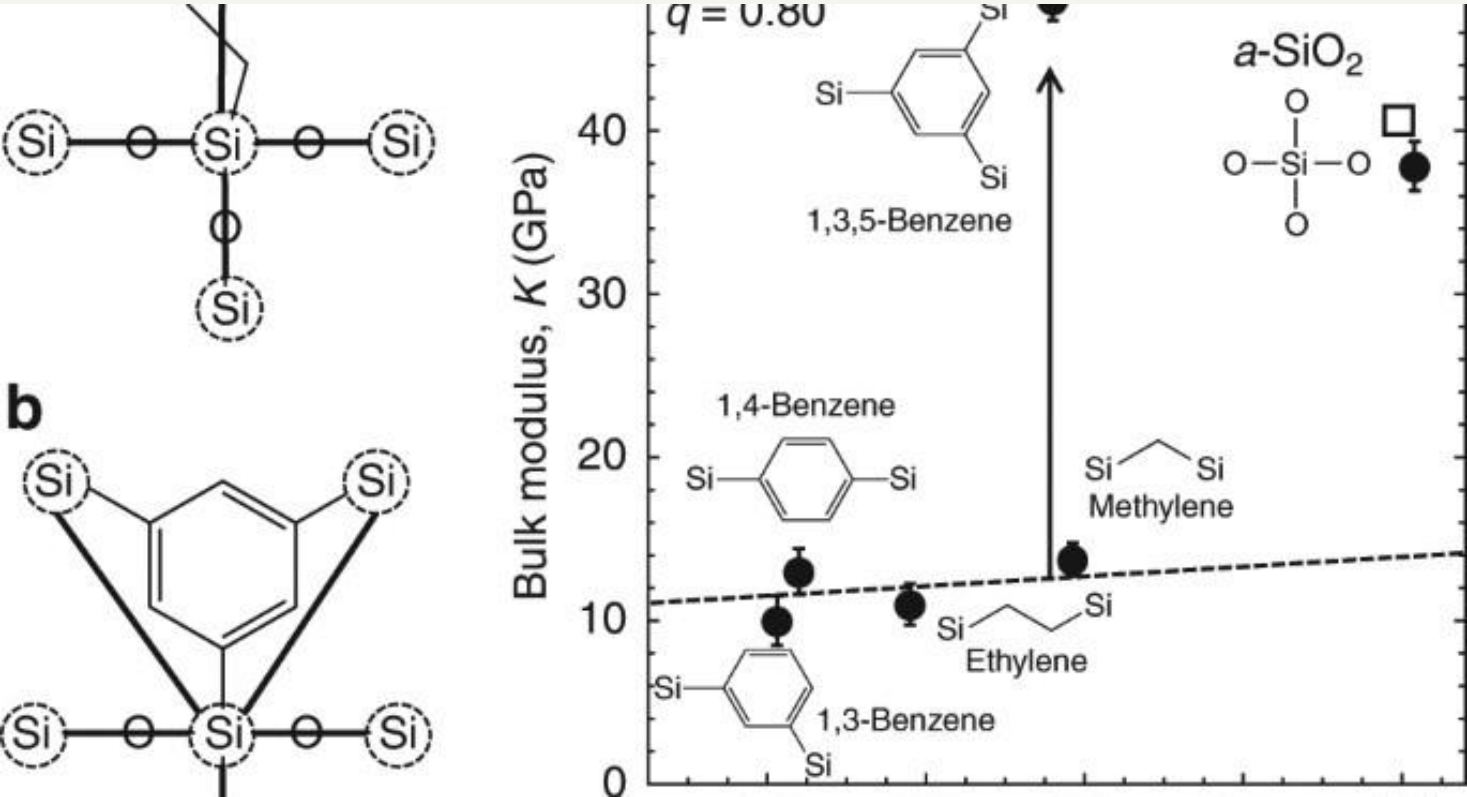


Fig. 1: Glass network structure and density correlation

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CHAPTER THREE

Characterization Techniques

Experimental and computational methods for comprehensive optical and radiation shielding evaluation.

Experimental & Computational Methods

Optical Characterization

■ **UV-Vis-NIR Spectrophotometer** (200-2500 nm):
Transmittance and absorbance spectra

■ **Abbe Refractometer**: Refractive index (n) measurement

■ **Tauc Plot**: Optical band gap (E_{opt}) calculation

■ **Urbach Energy** (ΔE): Structural disorder analysis

Structural Characterization

■ **X-ray Diffraction (XRD)**: Amorphous structure confirmation; no crystalline peaks

■ **FTIR Spectroscopy**: Functional groups and network connectivity

■ **Density**: Archimedes principle

■ **Molar Volume**: $V_m = M/\rho$ calculation

Radiation Shielding

■ **Phy-X/PSD**: LAC, MAC, HVL, MFP, Z_{eff} , N_{eff}

■ **MCNP/Geant4/FLUKA**: Monte Carlo validation

■ **Energy Range**: 0.015 - 15 MeV

■ **XCOM/WinXCom**: Cross-sectional data

Computational Simulation Tools

Software	Primary Function	Application
Phy-X/PSD	Online shielding parameter calculator	LAC, MAC, HVL, MFP, Zeff, Neff
MCNP-5 / MCNPX	Monte Carlo N-Particle transport	Gamma/neutron transport simulation
Geant4 / FLUKA	Toolkit for particle simulation	Photon interaction validation

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CHAPTER FOUR

Expected Outcomes

Anticipated results for optical transparency and radiation shielding performance of the developed glass compositions.

Expected Results & Performance Targets

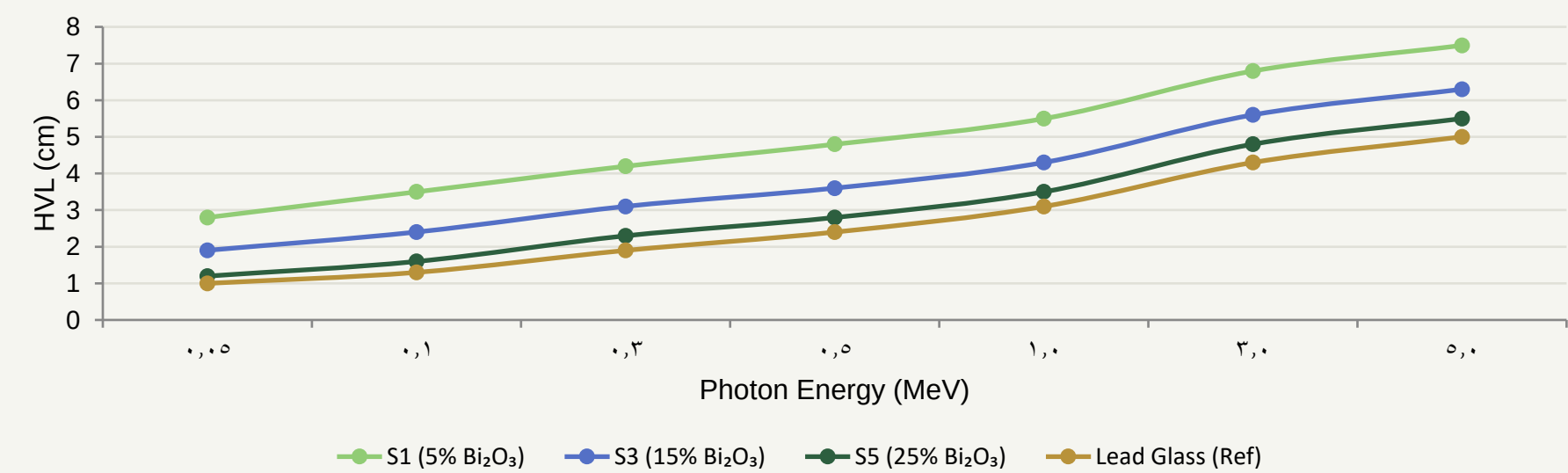
Optical Properties (Expected)

- **Visible light transmittance:** 70-85% for optimized compositions
- **Refractive index (n):** 1.6 - 2.0 range depending on heavy metal content
- **Optical band gap (E_{opt}):** Decreases with increasing heavy metal oxide content
- **Urbach energy (ΔE):** Increases with modifier concentration indicating structural changes

Radiation Shielding (Expected)

- **LAC:** Increases with density and Z_{eff} ; higher Bi_2O_3 = better attenuation
- **HVL:** Comparable to lead glass at equivalent energies for optimized samples
- **Z_{eff} :** > 40 expected for high-performance compositions
- **Best ratio:** 25-35 mol% Bi_2O_3 or Gd_2O_3 for shielding-to-transparency

Expected HVL vs. Photon Energy



BENCHMARK TARGETS

vs. Lead Glass: Achieve 80-95% of lead glass attenuation without toxicity

vs. Concrete: 2-3x better attenuation per unit thickness + transparency

vs. RS-520: Optimized compositions match or exceed commercial performance

Best: 70% Bi_2O_3 :30% SiO_2 — lowest HVL at 0.511 MeV

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CHAPTER FIVE

Theoretical Program

Fundamental equations and computational framework for calculating shielding parameters.

Theoretical Framework & Equations

FUNDAMENTAL LAW

Beer–Lambert Law: $I = I_0 e^{-\mu x}$ — where I_0 = incident intensity, μ = linear attenuation coefficient, x = material thickness

Linear Attenuation Coeff. (LAC)

$$\mu = \sum_i w_i (\mu/\rho)_i \text{ (cm}^{-1}\text{)}$$

w_i = weight fraction of component i

Mass Attenuation Coeff. (MAC)

$$\mu/\rho = \sum_i w_i (\mu/\rho)_i \text{ (cm}^2\text{/g)}$$

Mixture rule for composite materials

Half-Value Layer (HVL)

$$HVL = \ln(2)/\mu = 0.693/\mu \text{ (cm)}$$

Thickness reducing intensity by 50%

Tenth-Value Layer (TVL)

$$TVL = \ln(10)/\mu = 2.303/\mu \text{ (cm)}$$

Thickness reducing intensity by 90%

Mean Free Path (MFP)

$$MFP = 1/\mu \text{ (cm)}$$

Average photon travel distance

Effective Atomic Number (Z_{eff})

$$Z_{eff} = [\sum_i w_i Z_i^4]^{1/4}$$

Weighted average for photoelectric effect

EFFECTIVE ELECTRON DENSITY

$$N_{eff} = N_A \times (Z_{eff}/A_{eff}) \times \rho \text{ (electrons/cm}^3\text{)}$$

BUILDUP FACTORS

EBF (Exposure) and EABF (Energy Absorption) for broad-beam geometry

Photon Interaction Mechanisms

Photoelectric Effect ($E < 0.1 \text{ MeV}$): $LAC \propto Z^4/E^{3.5}$

Compton Scattering (0.1-5 MeV): Less Z-dependent

Pair Production ($E > 5 \text{ MeV}$): $LAC \propto Z^2$

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CHAPTER SIX

Nuclear Shielding Parameters

Detailed analysis of radiation attenuation metrics and comparative performance evaluation.

Shielding Parameters & Comparative Analysis

Table 2: Key Shielding Parameters at Selected Energies

Parameter	0.05 MeV	0.1 MeV	0.3 MeV	0.5 MeV	1.0 MeV	3.0 MeV	5.0 MeV
LAC (cm ⁻¹)	26.4	18.2	8.5	5.2	3.1	1.8	1.4
MAC (cm ² /g)	5.40	3.72	1.74	1.06	0.63	0.37	0.29
HVL (cm)	0.026	0.038	0.082	0.133	0.224	0.385	0.495
MFP (cm)	0.038	0.055	0.118	0.192	0.323	0.556	0.714
Z _{eff}	52.4	48.2	42.1	38.5	34.2	30.1	28.4

Energy-Dependent Interaction Regimes

Low Energy (E < 0.1 MeV)

Photoelectric effect dominates. $LAC \propto Z^4/E^{3.5}$. High-Z elements (Bi, W) provide maximum attenuation.

Medium Energy (0.1–5 MeV)

Compton scattering dominates. Less Z-dependence. Density becomes primary factor for attenuation.

High Energy (E > 5 MeV)

Pair production dominates. $LAC \propto Z^2$. High-Z glass compositions regain advantage.

Performance Comparison Summary



KEY FINDINGS

- 70% Bi₂O₃:30% SiO₂ shows lowest HVL at 0.511 MeV (PET energy)
- Lead-free compositions achieve 80–95% of lead glass attenuation
- Glass offers 2–3x better attenuation than concrete + transparency

References

- [1] Assessing the effectiveness of glass in radiation shielding, *Radiation Protection and Dosimetry*, Springer Nature, 2025. <https://doi.org/10.1007/s43939-025-00371-y>
- [2] Sadeq, M.S., et al., "Enhancing the gamma-ray attenuation parameters of mixed bismuth/barium borosilicate glasses using experimental method, Geant4 code and XCOM software," *Progress in Nuclear Energy*, 145, 104124, 2022.
- [3] Mustafa, I.S. & Sayyed, M.I., "Radiation parameterizations and optical characterizations for glass shielding composed of SLS waste glass and lead-free materials," *Nuclear Engineering and Technology*, 54, 4708, 2022. <https://doi.org/10.1016/j.net.2022.08.009>
- [4] Alqahtani, A.M., et al., "Study of ionizing radiation attenuation of glass as radiation shield," *Chalcogenide Letters*, 19(4), 227–238, 2022.
- [5] Sakar, E., et al., "Phy-X/PSD: development of a user friendly online software for calculation of parameters relevant to radiation shielding and dosimetry," *Radiation Physics and Chemistry*, 166, 108496, 2020.
- [6] Sayyed, M.I., et al., "Comparative investigations of gamma and neutron radiation shielding parameters for different borate and tellurite glass systems using WinXCom program and MCNPX code," *Materials Chemistry and Physics*, 215, 183–202, 2018.
- [7] Abouhaswa, A.S., et al., "Synthesis, structural, optical and radiation shielding features of tungsten trioxides doped borate glasses using Monte Carlo simulation and phy-X program," *Journal of Non-Crystalline Solids*, 543, 120134, 2020.
- [8] Issa, S.A.M., et al., "Evaluating the effectiveness of tellurium-molybdenum oxide glass systems for radiation shielding protection," *Multidisciplinary Materials Chronicles*, 1(1), 19–29, 2024.
- [9] Kassab, L.R.P., et al., "Gallium (III) oxide reinforced novel heavy metal oxide (HMO) glasses: a focusing study on synthesis, optical and gamma-ray shielding properties," *Ceramics International*, 48(10), 14261–14272, 2022.
- [10] Stalin, S., et al., "Influence of Bi₂O₃/WO₃ substitution on the optical, mechanical, chemical durability and gamma ray shielding properties of lithium-borate glasses," *Ceramics International*, 47(4), 5286–5299, 2021.

THANK YOU

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Questions & Discussion